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Subject : Software lab

Practical :5

Aim: Write a Java Application using JSP to create a Registration Form having attribute as name, password, email, phone no., gender, address, city, state, pin code. The form will allow users to submit their details into database. Upon form submission, login page should be displayed and email id and password should be verified from database. If user is valid then Welcome page should be displayed with session name.

Theory: Java Application Using JSP for Registration and Login

This application demonstrates the creation of a Registration Form using JSP (JavaServer Pages) and JDBC (Java Database Connectivity) to store user details in a database. Upon successful registration, the user is redirected to a Login Page, where their credentials are verified against the database. If valid, a Welcome Page is displayed with the user's session name.

Key Components:

1. JSP (JavaServer Pages):

- JSP is a server-side technology used to create dynamic web pages.
- It allows embedding Java code within HTML using special tags (<% %>).
- In this application, JSP is used to create the Registration Form, Login Form, and Welcome Page.

2. JDBC (Java Database Connectivity):

- JDBC is a Java API used to connect and execute queries with databases.

- It enables the application to interact with the MySQL database to store and retrieve user data.

3. Servlets:

- Servlets are Java classes that handle HTTP requests and responses.
- In this application, servlets are used to process form submissions, validate user credentials, and manage sessions.

4. Session Management:

- HTTP sessions are used to track user activity across multiple requests.
- The user's email is stored in the session upon successful login and displayed on the Welcome Page.

Code : DatabaseConnection.java

```
package Kunal;

import java.sql.Connection;
import java.sql.DriverManager;

public class DatabaseConnection {
    public static Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return
        DriverManager.getConnection("jdbc:mysql://localhost:3306/registrations",
        "root", "pass123");
    }
}
```

(LoginServlet.java)

```
package Kunal;

import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.ServletException;
```

```

import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        Connection conn = null;
        PreparedStatement stmt = null;
        ResultSet rs = null;

        try {
            conn = DatabaseConnection.getConnection();
            String sql = "SELECT * FROM users_demo WHERE email=? AND
password=?";
            stmt = conn.prepareStatement(sql);
            stmt.setString(1, request.getParameter("email"));
            stmt.setString(2, request.getParameter("password"));
            rs = stmt.executeQuery();

            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("user", rs.getString("name"));
                response.sendRedirect("welcome.jsp");
            } else {
                response.sendRedirect("login.jsp?message=Try+again:+your+email+or
password+is+incorrect");
            }
        } catch (Exception e) {
            e.printStackTrace();
            response.sendRedirect("login.jsp?message=Login+Failed");
        } finally {

```

```

        try {
            if (rs != null) rs.close();
            if (stmt != null) stmt.close();
            if (conn != null) conn.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
}

```

RegisterServlet.java

```

package Kunal;
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/RegisterServlet")
public class RegisterServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        Connection conn = null;
        PreparedStatement stmt = null;

        try {
            conn = DatabaseConnection.getConnection();
            String sql = "INSERT INTO users_demo (name, email, password, phone,
gender, address, city, state, pin_code) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)";
            stmt = conn.prepareStatement(sql);

```

```

        stmt.setString(1, request.getParameter("name"));
        stmt.setString(2, request.getParameter("email"));
        stmt.setString(3, request.getParameter("password")); // Storing plain text
password
        stmt.setString(4, request.getParameter("phone"));
        stmt.setString(5, request.getParameter("gender"));
        stmt.setString(6, request.getParameter("address"));
        stmt.setString(7, request.getParameter("city"));
        stmt.setString(8, request.getParameter("state"));
        stmt.setString(9, request.getParameter("pin_code"));

        stmt.executeUpdate();
        response.sendRedirect("login.jsp"); // Redirect to login page

    } catch (Exception e) {
        e.printStackTrace();
        response.sendRedirect("error.jsp?message=Registration Failed");
    } finally {
        try {
            if (stmt != null) stmt.close();
            if (conn != null) conn.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
}

```

LogoutServlet.java

```

package Kunal;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;

```

```
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
```

```
@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession();
        session.invalidate();
        response.sendRedirect("login.jsp");
    }
}
```

```
// WEB-INF
```

index.jsp contains (Registration form)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Register</title>
    <style>
        /* Google Font */
        @import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap');

        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: 'Poppins', sans-serif;
        }
```

```
body {  
  background: linear-gradient(to right, #ff9966, #ff5e62);  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  height: 100vh;  
  padding: 20px;  
}
```

```
.container {  
  background: white;  
  padding: 30px;  
  border-radius: 12px;  
  box-shadow: 0px 10px 30px rgba(0, 0, 0, 0.2);  
  width: 400px;  
  text-align: center;  
}
```

```
.container h2 {  
  margin-bottom: 15px;  
  font-size: 24px;  
  font-weight: 600;  
  color: #333;  
}
```

```
input, textarea, select {  
  width: 100%;  
  padding: 10px;  
  margin: 10px 0;  
  border: 1px solid #ccc;  
  border-radius: 6px;  
  font-size: 14px;  
}
```

```
input:focus, textarea:focus {
```

```
    outline: none;
    border-color: #ff5e62;
    box-shadow: 0px 0px 5px rgba(255, 94, 98, 0.5);
}
```

```
.gender-label {
    display: flex;
    justify-content: center;
    gap: 15px;
    margin-bottom: 10px;
}
```

```
.btn {
    background: #ff5e62;
    color: white;
    padding: 10px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-size: 16px;
    transition: background 0.3s;
    width: 100%;
}
```

```
.btn:hover {
    background: #e84142;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
  <h2>Register</h2>
```

```
  <form action="RegisterServlet" method="post">
```



```

        <input type="text" name="name" placeholder="Enter Name" required>
        <input type="email" name="email" placeholder="Enter Email" required>
        <input type="password" name="password" placeholder="Enter Password"
required>
        <input type="text" name="phone" placeholder="Enter Phone No." required>

        <div class="gender-label">
            <label><input type="radio" name="gender" value="Male" required>
Male</label>
            <label><input type="radio" name="gender" value="Female" required>
Female</label>
        </div>

        <textarea name="address" placeholder="Enter Address"
required></textarea>
        <input type="text" name="city" placeholder="Enter City" required>
        <input type="text" name="state" placeholder="Enter State" required>
        <input type="text" name="pin_code" placeholder="Enter Pin Code"
required>

        <input type="submit" value="Register" class="btn">
        <p>Already have an account?<a href="login.jsp">Login</a></p>
    </form>
</div>

</body>
</html>

```

Login.jsp

```

<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%
    String errorMessage = request.getParameter("error");
%>
<!DOCTYPE html>
<html lang="en">

```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login</title>
  <style>
    /* Apply Google Font */
    @import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap');

    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Poppins', sans-serif;
    }

    body {
      background: linear-gradient(to right, #4facfe, #00f2fe);
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .container {
      background: white;
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0px 10px 30px rgba(0, 0, 0, 0.2);
      width: 350px;
      text-align: center;
    }

    .container h2 {
```

```
margin-bottom: 15px;
font-size: 24px;
font-weight: 600;
color: #333;
}
```

```
.error-message {
  color: red;
  font-size: 14px;
  margin-bottom: 10px;
}
```

```
input {
  width: 100%;
  padding: 10px;
  margin: 10px 0;
  border: 1px solid #ccc;
  border-radius: 6px;
  font-size: 14px;
}
```

```
input:focus {
  outline: none;
  border-color: #4facfe;
  box-shadow: 0px 0px 5px rgba(79, 172, 254, 0.5);
}
```

```
.btn {
  background: #4facfe;
  color: white;
  padding: 10px;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-size: 16px;
```

```

        transition: background 0.3s;
        width: 100%;
    }

    .btn:hover {
        background: #007bff;
    }
</style>
</head>
<body>

<div class="container">
    <h2>Login</h2>
    <%
        String message = request.getParameter("message");
        if (message != null) {
    %>
    <p style="color:red;"><%= message %></p>
    <%
        }
    %>

    <form action="LoginServlet" method="post">
        <input type="email" name="email" placeholder="Enter Email" required>
        <input type="password" name="password" placeholder="Enter Password"
required>
        <input type="submit" value="Login" class="btn">
        <p>Don't have an account ? <a href="index.jsp">SignUp</a></p>
    </form>
</div>

</body>
</html>

```

Welcome.jsp

```
<%@ page import="javax.servlet.http.HttpSession" %>
<%
    HttpSession sessionObj = request.getSession(false);
    if (sessionObj == null || sessionObj.getAttribute("user") == null) {
        response.sendRedirect("login.jsp");
    }
%>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Welcome</title>
    <style>
        /* Google Font */
        @import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap');

        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: 'Poppins', sans-serif;
        }

        body {
            background: linear-gradient(to right, #36d1dc, #5b86e5);
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            text-align: center;
```

```
padding: 20px;  
}
```

```
.container {  
  background: white;  
  padding: 30px;  
  border-radius: 12px;  
  box-shadow: 0px 10px 30px rgba(0, 0, 0, 0.2);  
  height: 200px;  
  width: 700px;  
  text-align: center;  
}
```

```
.container h2 {  
  font-size: 26px;  
  font-weight: 600;  
  color: #333;  
  margin-bottom: 15px;  
}
```

```
.user-name {  
  font-size: 22px;  
  font-weight: bold;  
  color: #5b86e5;  
}
```

```
.logout-btn {  
  background: #ff5e62;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
  border-radius: 6px;  
  cursor: pointer;  
  font-size: 16px;  
  text-decoration: none;
```

```
display: inline-block;
margin-top: 15px;
transition: background 0.3s ease-in-out;
}
```

```
.logout-btn:hover {
    background: #e84142;
}
```

```
</style>
</head>
<body>

<div class="container">
    <h2>Welcome, <span class="user-name"><%= sessionObj.getAttribute("user")
%></span>!</h2>
    <p>Welcome to our community! Together, we grow</p>
    <a href="LogoutServlet" class="logout-btn">Logout</a>
</div>
</body>
</html>
```

Output:

Register

Enter Name

Enter Email

Enter Password

Enter Phone No.

☐ Male ☐ Female

Enter Address

Enter City

Enter State

Enter Pin Code

Register

Already have an account? [Login](#)

Login

Enter Email

Enter Password

Login

Don't have an account ? [SignUp](#)

Database:

```
MySQL 8.0 Command Line Cli x + v

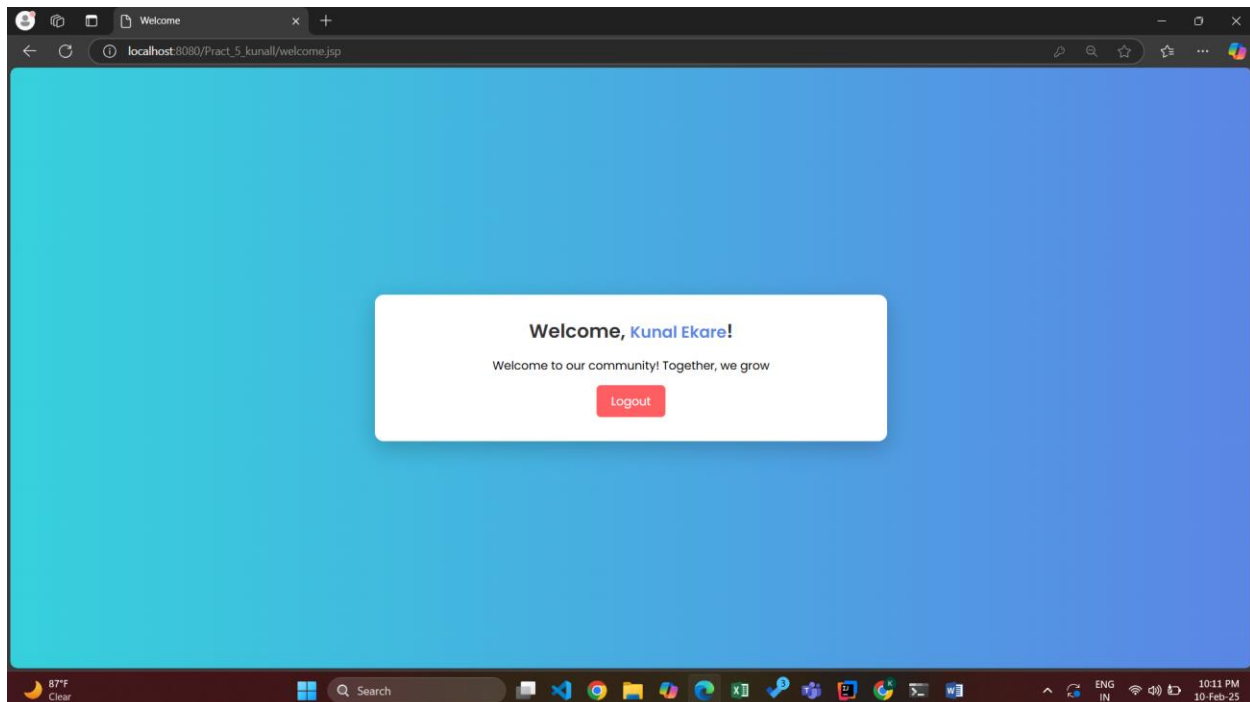
mysql> select * from users_demo;
Empty set (0.01 sec)

mysql> use Registration;
Database changed
mysql> CREATE TABLE users_demo (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> name VARCHAR(100),
-> email VARCHAR(100) UNIQUE,
-> password VARCHAR(255),
-> phone VARCHAR(15),
-> gender VARCHAR(10),
-> address TEXT,
-> city VARCHAR(50),
-> state VARCHAR(50),
-> pin_code VARCHAR(10)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> select * from users_demo;
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | name   | email                | password | phone   | gender | address   | city     | state   | pin_code |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1  | Kunal Ekare | kunalekare02@gmail.com | Pass@1234p | 9322433172 | Male   | Urga Nagar | chandrapur | Maharastra | 442906 |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select * from users_demo;
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | name   | email                | password | phone   | gender | address   | city     | state   | pin_code |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1  | Kunal Ekare | kunalekare02@gmail.com | Pass@1234p | 9322433172 | Male   | Urga Nagar | chandrapur | Maharastra | 442906 |
| 2  | Rahul Sharma | rahul.sharma@email.com | rahul123  | 9876543210 | Male   | 12, MG Road | Mumbai   | Maharashtra | 400001 |
| 3  | Priya Verma | priya.verma@email.com | priya456  | 8765432109 | Female | 22B, Green Park | Delhi    | Delhi       | 110016 |
| 4  | Amit Gupta | amit.gupta@email.com | amit789   | 7654321098 | Male   | 45, Lakeview Street | Kolkata  | West Bengal | 700001 |
| 5  | Anjali Das | anjali.das@email.com | anjali888 | 4321098765 | Female | 32, Marine Drive | Chennai  | Tamil Nadu  | 600001 |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```



Conclusion:

From this practical, we learned how to develop a **Java web application using JSP and Servlets** for user registration and login. We implemented:

- **Form Handling** – Capturing user details through an HTML form and submitting data to a servlet.
- **Database Connectivity** – Storing and retrieving user information using JDBC.
- **User Authentication** – Verifying login credentials against the database
- **Session Management** – Maintaining user sessions to personalize the welcome page.
- **Error Handling** – Displaying appropriate messages for incorrect login attempts.

This practical enhanced our understanding of **JSP, Servlets, JDBC, and session management**, which are essential for building dynamic web applications.